

Project Design Phase-II Phase: Technology Stack (Architecture & Stack)

Date	7 July 2026
Team ID	
Project Name	Nutrition Assistant
Maximum Marks	4 Marks

1. Technical Architecture Overview

The application features a modern 3-Tier architecture specifically chosen for scalability, security, and developer maintainability. Presentation is handled by React 19 and Vite; API servers are developed with Node.js and Express.js, and data persistence utilizes MongoDB Atlas. External cloud services integrated include Gemini AI and Cloudinary CDN.

2. Table-1 : Components & Technologies

S.No	Component	Description	Technology Stack
1.	User Interface	Web client layout designed for mobile/desktop. Uses Tailwind CSS and Framer Motion.	React.js 19, Vite, Tailwind v4
2.	Application Logic-1	Authentication, user profiles, and CRUD trackers backend route handlers.	Node.js, Express.js
3.	Application Logic-2	AI Nutrition Concierge chatbot triggers processing natural language questions.	Gemini AI API SDK
4.	Application Logic-3	Image processing, avatar photos, and food receipt storage system logic.	Cloudinary SDK Wrapper
5.	Database Schema	Strict object modeling definitions mapping logged collections.	Mongoose Schemas
6.	Cloud Database	NoSQL document persistence system deployed in the cloud.	MongoDB Atlas
7.	File Storage	Online CDN server managing profile uploads and photo streams.	Cloudinary Service CDN
8.	External API-1	External generative model responding to wellness and macro requests.	Google Gemini API
9.	External API-2	External ingredients information source API lookup queries.	Edamam / USDA Foods API
10.	Machine Learning Model	Contextual engine assessing nutrient contents inside food prompt strings.	Gemini 1.5 Flash
11.	Infrastructure	Automated deployment hosting configurations.	Vercel (Client), Render (API)

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3. Table-2 : Application Characteristics

S.No	Characteristics	Description	Technology Used
1.	Open-Source Frameworks	Standard development stack frameworks utilizing npm integrations.	React 19, Express.js, Mongoose, Axios, Tailwind CSS, Framer Motion
2.	Security Implementations	Secure session tokens, input sanitation, validation libraries, and hashed details.	JWT, Bcrypt hashing, Zod validations, dotenv keys, CORS middleware config
3.	Scalable Architecture	Separation of concerns ensuring clear client, controller, and schema logic layers.	3-Tier MVC architecture separating controllers from Mongoose models
4.	Availability Specifications	Cloud-hosted services handling traffic routing and connection queries.	MongoDB Atlas multi-region replicas, Vercel/Render serverless infrastructure
5.	Performance Practices	Latency optimization methods.	Database indexing, static bundles compression, structured lightweight responses